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IT School: AI - Epoch 1

Python Fundamentals

Iteration 1

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Agenda

01. Why Python?

02. Setup

03. Variables & Data Types

04. Operators

05. Control Flow

06. Data Structures

07. Functions

08. Built-in functions

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01. Why Python?

Easy & Readable:

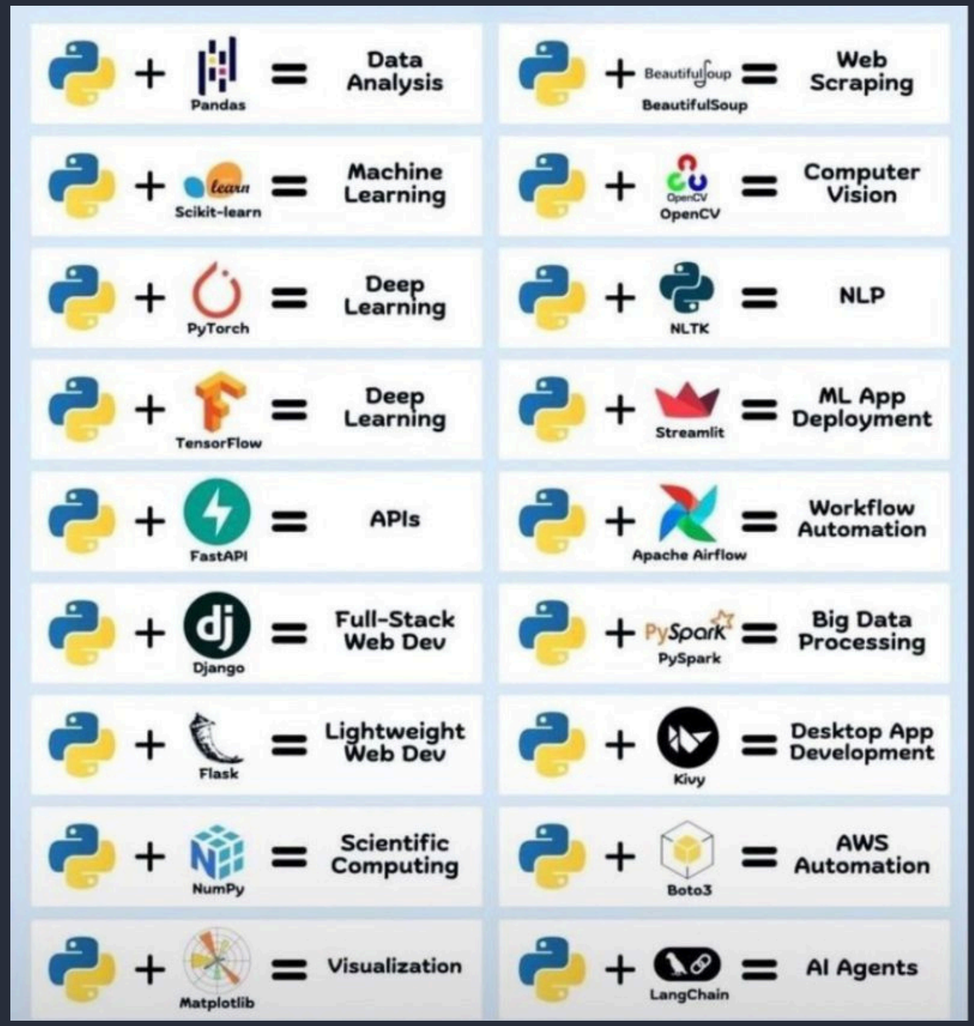
- 👉 Clean syntax that reads like English
- 👉 Less time fighting with code, more time solving problems
- 👉 Fast to learn, fast to prototype

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01. Why Python?

Powerful Ecosystem:

- NumPy, Pandas -> data manipulation
- Matplotlib, Seaborn -> visualization
- TensorFlow, PyTorch -> deep learning
- scikit-learn -> machine learning



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02. Setup

- Install **Python**: https://realpython.com/installing-python/
- Install **VS Code** (or your preferred IDE): https://code.visualstudio.com/docs/setup/setup-overview
- Install **Jupyter Notebook** extension in VS Code: https://code.visualstudio.com/docs/datascience/jupyter-notebooks



- Alternatives (no installation needed):
  - Google Colab: https://colab.research.google.com
  - Kaggle Notebooks: https://www.kaggle.com/code

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Coding Section

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Additional Resources

📌 HackerRank’s Python Challenges: https://www.hackerrank.com/domains/python

📌 Harvard’s Introduction to Programming with Python: https://cs50.harvard.edu/python/

📌 And always ✨ read the docs ✨: https://docs.python.org/3/

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Any Questions ?

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Thank you for your attention ✨

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